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Operations Research

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Sample solution to the demo tasks 7

Demo Exercises

Exercise D7. 1. *Cutting*

Problem

A carpentry shop has $20m$ and $17m$ boards at its disposal. It receives an order from a customer to produce 30 boards of $6m$ length (type A), 15 boards of $7m$ length (type B) and 40 boards of $3m$ length (type C). The use of a $20m$ board costs 20 euros, and the use of a $17m$ board costs 17 euros. The joinery wants to fulfill the order in such a way that the costs for it are minimal. It would like to solve this problem using column generation.

a) For the CG algorithm, an optimal cutting pattern must be generated in each iteration with respect to a given utility vector $u = (u_A, u_B, u_C) \in \mathbb{R}^3$ and a given length $l \in \{17, 20\}$ of the initial board. Let u_T denote the utility of a board of type $T \in \{A, B, C\}$. Formulate an IP that solves the problem "find an optimal cut with respect to u and l".

b) We choose as (bad) initial basis the 3 cutting patterns where exactly 1 board of one type is cut from a board of length $17m$. Run one iteration of the CG algorithm with this basis.

Solution

a) Given u and l we have to solve the following problem:

$$\begin{aligned} \max\ & u_A a_A + u_B a_B + u_C a_C - c_T \\ s.t.\ & 6a_A + 7a_B + 3a_C \leq l \\ & a_A, a_B, a_C \in \mathbb{N}_0 \end{aligned}$$

b) Let x_i be the number of boards cut according to pattern $i = 1, 2, 3$: At cutting pattern $i = 1$ one board of type A is cut out, at $i = 2$ one of type B and at $i = 3$ one of type C. The corresponding inequalities are therefore

$$\begin{aligned} x_1 &\geq 30 \\ x_2 &\geq 15 \\ x_3 &\geq 40 \end{aligned}$$

Consequently, B_0 is given by

$$B_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

und

$$b = \begin{pmatrix} 30 \\ 15 \\ 40 \end{pmatrix}.$$

The costs are given by $17x_1 + 17x_2 + 17x_3$, so that

$$c_{B_0} = \begin{pmatrix} 17 \\ 17 \\ 17 \end{pmatrix}.$$

So, it holds that

$$c_{B_0}^T B_0^{-1} = (17, 17, 17).$$

Let $A^0 = (a_A^0, a_B^0, a_C^0)$ be the cutting pattern, which maximizes the non-basis variables, i.e.,

$$(17, 17, 17)A^0 - c_{A^0}.$$

Since we don't yet know if the optimal pattern requires a 17- or 20-meter board, we need to solve the following two IPs

$$\begin{aligned} \max 17a_A^0 + 17a_B^0 + 17a_C^0 - 17 \\ s.t. 6a_A^0 + 7a_B^0 + 3a_C^0 \leq 17 \\ a_A, a_B, a_C \in \mathbb{N}_0 \end{aligned}$$

and

$$\begin{aligned} \max 17a_A^0 + 17a_B^0 + 17a_C^0 - 20 \\ s.t. 6a_A^0 + 7a_B^0 + 3a_C^0 \leq 20 \\ a_A, a_B, a_C \in \mathbb{N}_0 \end{aligned}$$

and choose the solutions with larger optimal value of the two. For $l = 17$ an optimal solution is given by $a_A^0 = a_B^0 = 0, a_C^0 = 5$ with function value 68. For $l = 20$ an optimal solution is given by $a_A^0 = a_B^0 = 0, a_0^C = 6$ with function value 82. So we add to our variables the pattern x_4 where 6 boards of length $3m$ are cut from a $20m$ board. Since only $a_0^C > 0$, the outgoing base variable is x_3. This completes the first iteration.
We get the new basis

$$B_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}.$$

The cost vector c_{B_1} is given by $c_{B_1} = (17, 17, 20)$.

Exercise D7. 2. *Vacation*

Problem

After an exhausting semester, the student Georgina plans to go on vacation with all her friends. In total, m people are going on vacation, with the i-th person wanting to bring $w_i \in \mathbb{N}$ kilograms of luggage. They plan to undertake the trip with several rental cars, where each person wants to travel in the car that transports their luggage, and the total weight of the luggage in the cars must not exceed $W \in \mathbb{N}$ kilograms each. Unfortunately, some of the friends have quarrels with each other, so they don't want to travel in the same car. Let $S \subseteq \{\{i,j\} \subseteq \{1,\ldots,m\},:,i \neq j\}$ denote the set of all quarrelsome pairs. Thus, $\{i,j\} \in S$ if and only if i does not want to sit in a car with j.

Of course, Georgina wants to rent as few cars as possible. She wants to approximate the optimal solution using Column Generation. To do this, she first formulates the following IP:

$$\begin{aligned} \min, \sum_{i=1}^{n} c_i x_i \\ s.t. Ax = b \\ x \in \{0,1\}^n. \end{aligned}$$

The matrix A has m rows and n columns, with each column describing a permissible "filling pattern" for a car. Furthermore, $c = (1,\ldots,1)^T \in \mathbb{R}^n$ and $b = (1,\ldots,1)^T \in \mathbb{R}^m$. The column $a \in \{0,1\}^m$ describes a car in which exactly those persons j with $a_j = 1$ are sitting. $x_i = 1$ means that the filling pattern in the i-th column is used. The filling pattern must comply with the rules stated above. The allowable total weight W of the car must not be exceeded, and for no pair $\{i,j\} \in S$ should $a_i = 1$ and $a_j = 1$ hold. Furthermore, at least one person must travel in each used car. We call such a filling pattern *valid*.

a) Describe the condition "$a = (a_1,\ldots,a_m)^T \in \{0,1\}^m$ is valid" through linear (in)equality constraints.

b) In this subtask, we assume that $w_i = 10$ for all $i \in \{1,\ldots,m\}$, $W = 40$, and $m \in \mathbb{N}$ is arbitrary. Specify the number n of columns of A for the two sets $S_1 = \emptyset$ and $S_2 = \{\{i,j\},:,i \neq j\}$ of quarrelsome friends.

c) Under what condition on the number $|S|$ of quarrelsome pairs is it particularly advisable to use the Column Generation Algorithm? Use the results from subtask (b) to justify your answer.

d) For this subtask, we assume that $m = 4$ people are going on vacation, $w_i = 10$, $W = 40$, and $S = \{\{1,2\},\{2,3\}\}$. Perform one iteration of the CG algorithm, starting with the basis in which each person travels alone in a car. Specify in particular the pricing problem and - with justification - the entering and exiting basic variables. Intermediate calculations need not be provided.

Solution

a) At least one person in the car:

$$\sum_{i=1}^{m} a_i \geq 1$$

Total weight is not exceeded:

$$\sum_{i=1}^{m} a_i w_i \leq W$$

Quarrelsome people not in the same car:

$$a_i + a_j \leq 1 \, \forall \{i, j\} \in S$$

b) S_1: A filling pattern is valid if and only if at least one and at most 4 people are in the car. Thus, only the vectors $a \in 0, 1^m$ that have at least one and at most 4 entries of 1 describe valid filling patterns. Therefore, A has

$$\binom{m}{4} + \binom{m}{3} + \binom{m}{2} + \binom{m}{1} \sim m^4$$

columns.
S_2: Since every person is quarrelsome with every other person, the only valid filling patterns are those where each person sits in their own car, i.e., the columns $a = e_i$ for $i \in \{1, \dots, m\}$. That is exactly $n = m$ columns.

c) If $|S|$ is small, there tend to be more filling patterns according to (b) than if $|S|$ is large. The use of the CG algorithm is particularly worthwhile when the number of columns significantly exceeds the number of rows of A. Therefore, the CG algorithm is particularly suitable here for small $|S|$.

d) The basis matrix is the identity matrix I_4. Since $c = (1, \dots, 1)$, $c_B^T B^{-1} = (1, 1, 1, 1)$. Thus, the pricing problem is

$$\begin{aligned} \max a_1 + a_2 + a_3 + a_4 - 1 \\ s.t.\, a_1 + a_2 + a_3 + a_4 \geq 1 \\ 10a_1 + 10a_2 + 10a_3 + 10a_4 \leq 40 \\ a_1 + a_2 \leq 1 \\ a_2 + a_3 \leq 1 \\ a_i \in \{0, 1\} \end{aligned}$$

The optimal solution for this is $a_1 = a_3 = a_4 = 1$, $a_2 = 0$ with value $3 - 1 = 2 > 0$. Thus, the variable corresponding to the column $a^* = (1, 0, 1, 1)^T$ enters the basis. Since $B^{-1}a^* = (1, 0, 1, 1)^T$, $B^{-1}b/B^{-1}a^* = (1, \infty, 1, 1)$. Therefore, we can replace any one of the three filling patterns $(1, 0, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$ with $(1, 0, 1, 1)$.